crawling: the operation is only performed when we have a "miss", while "hits" usually triggers a more in-depth comparison (for instance, on a hit, retrieving from disk just the first few lines or the first block of a document, and comparing them).

4.7.5 Spell checker

Simpler versions of spell checkers used to employ Bloom filters as dictionaries. For every word of the text examined, a lookup on a Bloom filter would validate the word as correct or mark it as a spelling error. The false positive occurrences would of course cause some spelling error to go undetected, but the odds of this happening could be controlled in advance. Today, however, spell checkers mostly takes advantage of *tries*: these data structures provide good performance on text searches without the false positives.

4.7.6 Distributed Databases and File Systems

Cassandra uses Bloom filters for index scans, to determine whether an *SSTable* has data for a particular row.

Likewise, Apache *HBase* uses Bloom filters as an efficient mechanism to test whether a *StoreFile* contains a specific row or row-col cell. This in turn boosts the overall read speed, by filtering out unnecessary disk reads of *HFile* blocks that don’t contain a particular row or row-column.